Snakemake preprocessing pipeline

Raw EEG Data

95 participants
36 stimuli · 14 ch · 256 Hz

`0_total_data.csv` · `0_mean_sd.csv`

1. Isolate Stimuli

from raw recording

0_isolate_stimuli.py

`3_fft_data.pck`

2. Extract FFT Spectral Features

2_isolate_emotiv_fft.py

`1_quality_filter.csv`

3. Inspect Signal Quality

1_inspect_quality.py

`2_filtered.csv`

4. Filter Samples

quality threshold

filter_samples.py

`4_normalized.csv`

5. Normalize Stimuli

z-score · IQR · baseline

3_normalize_stimuli.py

`5_features.csv`

6. Feature Extraction

4_perform_feature_extract.py

`EEG_LSTM_windows/` · `*.pkl`

7. Prepare Windows for Training

512 samples · 2 s

5_make_data_for_LSTM.py

Model-ready Dataset

shape (512, 14)

Snakemake 7.32.4 · Python 3.12

per-subject parallelism